

Amarsh Pedapati

Senior AI Engineer | Agentic Systems & LLM Orchestration

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Nationality: Indian | Notice Period: 30 days

Senior AI Engineer (IIT Hyderabad) with 3+ years building and maintaining production agentic systems, RAG pipelines, and LLM infrastructure end-to-end. Shipped an enterprise AI Cockpit to beta, a 12x-faster CI/CD migration agent, and evaluation-driven RAG systems with full observability. Works autonomously across the stack, from LLM design to FastAPI backend to stakeholder-facing UI.

EDUCATION

Indian Institute of Technology (IIT), Hyderabad

Jul 2019 – May 2023

Bachelor of Technology, Artificial Intelligence

WORK EXPERIENCE

Senior AI Engineer | OpenText, Hyderabad

Jan 2026 – Present

AI Cockpit: Enterprise Agent Orchestration & MCP Integration

- Reduced project planning and test automation from **2 weeks to 2 days** by architecting an enterprise AI Cockpit for OpenText clients; led a 4-engineer team to deliver, now in beta with enterprise clients.
- Designed the agentic architecture with a master agent routing natural-language requests through a dynamic agent registry via **A2A protocol** to domain Expert Agents, each grounded by **RAG** over product documentation and historical examples.
- Built production-grade guardrails, role-based authorization, and structured-output validation across all agents; **MCP** tool integrations enable direct action across enterprise systems including SAP HANA; served via **FastAPI** with end-to-end **LangSmith** observability.

CI/CD Migration Agent & DevOps Copilot Plugin

- Achieved **12x acceleration** on a **770-job** Jenkins-to-GitLab migration, compressing a 12-month effort to under 1 month by architecting a **LangGraph**-based self-healing multi-agent system with persistent state and hard HITL checkpoints; won company hackathon among ~200 participants and drove adoption across **10+ teams**.
- Shipped an IDE-agnostic DevOps Copilot plugin exposing composable agent skills via **MCP-compatible** tool interfaces, enabling non-DevOps engineers to generate GitLab CI workflows and execute migration steps from agentic development environments; led cross-functional collaboration with DevOps and QA teams to map manual workflows before automating them end-to-end.

Data Scientist | Blend360, Hyderabad

Oct 2023 – Dec 2025

For Visa: GenAI Model Explainability Platform

- Designed, built, and maintained a full-stack GenAI explainability platform for Visa, converting raw model discrepancy signals into evidence-backed executive narratives; shipped a client-facing UI for non-technical stakeholders including C-suite and department heads, collaborating cross-country with US-based teams.
- Engineered an agentic chatbot for **RAG**-backed data exploration and built a production evaluation pipeline using **RAGAS** to measure Answer Relevancy, Faithfulness, and Hallucination risk; owned post-launch maintenance, iteration, and quality improvements across both systems.

For Walmart: Holiday Campaign (Data Engineering & ML)

- Achieved **45% uplift** in top decile and enabled targeting of **13.4M customers** from a 130M customer base by building end-to-end segmentation and uplift modeling pipelines with **Scikit-learn** and probability calibration for Walmart's holiday campaign.
- Engineered a high-performance **Spark** datamart on **Snowflake**, delivering **7x faster** query performance and **1000x ETL runtime improvement**; improved model lift by **100%** through feature engineering and calibration.

ML Engineer Intern | AIBOD, Fukuoka, Japan

Jan – Jun 2022

- Reduced misclassification rate by **32%** and improved accuracy by **18%** by building a **PyTorch** computer vision pipeline with out-of-distribution detection and uncertainty estimation for an AI-powered unmanned retail system at a Japanese AI startup.

PROJECTS

PersonaRAG: Eval-Driven Hybrid RAG System | [GitHub](#)

- Built an independent eval-driven RAG research project to test hallucination control at scale; designed a hybrid retrieval system over **4.75M words** with multi-query expansion, cross-encoder reranking, and a custom faithfulness guard; achieved **0 false positives** across 30+ adversarial probes and **37/37** pass rate across independent eval suites using an **LLM-as-Judge** framework.

TECHNICAL SKILLS

Agentic AI & LLMs: Multi-Agent Systems, LangGraph, LangChain, MCP, A2A Protocol, RAG, Hybrid Retrieval, Multi-query Retrieval, Cross-encoder Reranking, Prompt Engineering, Agentic Workflows, Guardrails, Hallucination Detection, LLM Evaluation, LLM-as-Judge, RAGAS, Embedding Models

ML & Frameworks: PyTorch, Scikit-learn, Hugging Face Transformers, PydanticAI, OpenAI SDK, Computer Vision, OOD Detection, Uncertainty Estimation

Backend, Infra & Cloud: FastAPI, REST APIs, Docker, Kubernetes, AWS Bedrock, AWS SageMaker, PostgreSQL, Redis, Snowflake, FAISS, Pinecone, ChromaDB, LangSmith, CI/CD, GitLab CI

